

IBM Data Analyst Capstone Project

Analyzing Technology Trends Using the Stack Overflow Developer Survey

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- **Date:** 29 June 2026



OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
- Appendix



EXECUTIVE SUMMARY



- This project analyzed the **Stack Overflow Developer Survey** to identify current and emerging technology trends among software developers.
- Collected and prepared survey data from multiple sources.
- Cleaned and transformed the dataset for analysis.
- Performed exploratory data analysis to identify technology trends.
- Created visualizations and interactive dashboards using IBM Cognos Analytics.
- Generated insights on programming languages, databases, platforms, web frameworks, and developer demographics.



INTRODUCTION

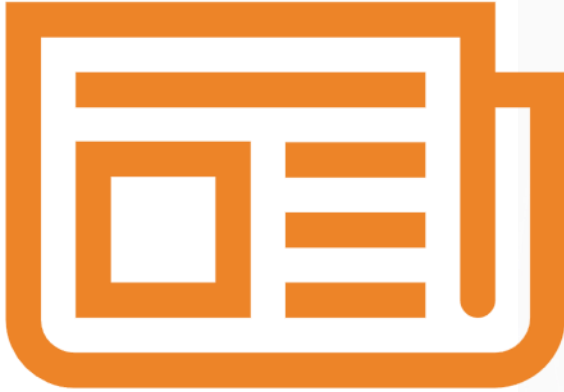


Purpose of the report, Target Audience & Value of the analysis

- **Purpose:** Analyze the Stack Overflow Developer Survey to identify current and future technology trends.
- **Dataset:** Stack Overflow Developer Survey responses from software developers worldwide.
- **Target Audience:** Students, aspiring data analysts, developers, recruiters, and technology decision-makers.
- **Value of the Analysis:**
 - Identify the most widely used technologies.
 - Understand future technology preferences among developers.
 - Explore developer demographics and professional backgrounds.



METHODOLOGY

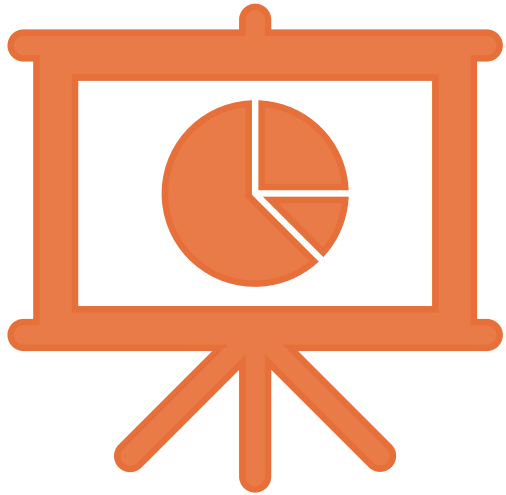


Data Sources, Data Collection & Data Wrangling Steps

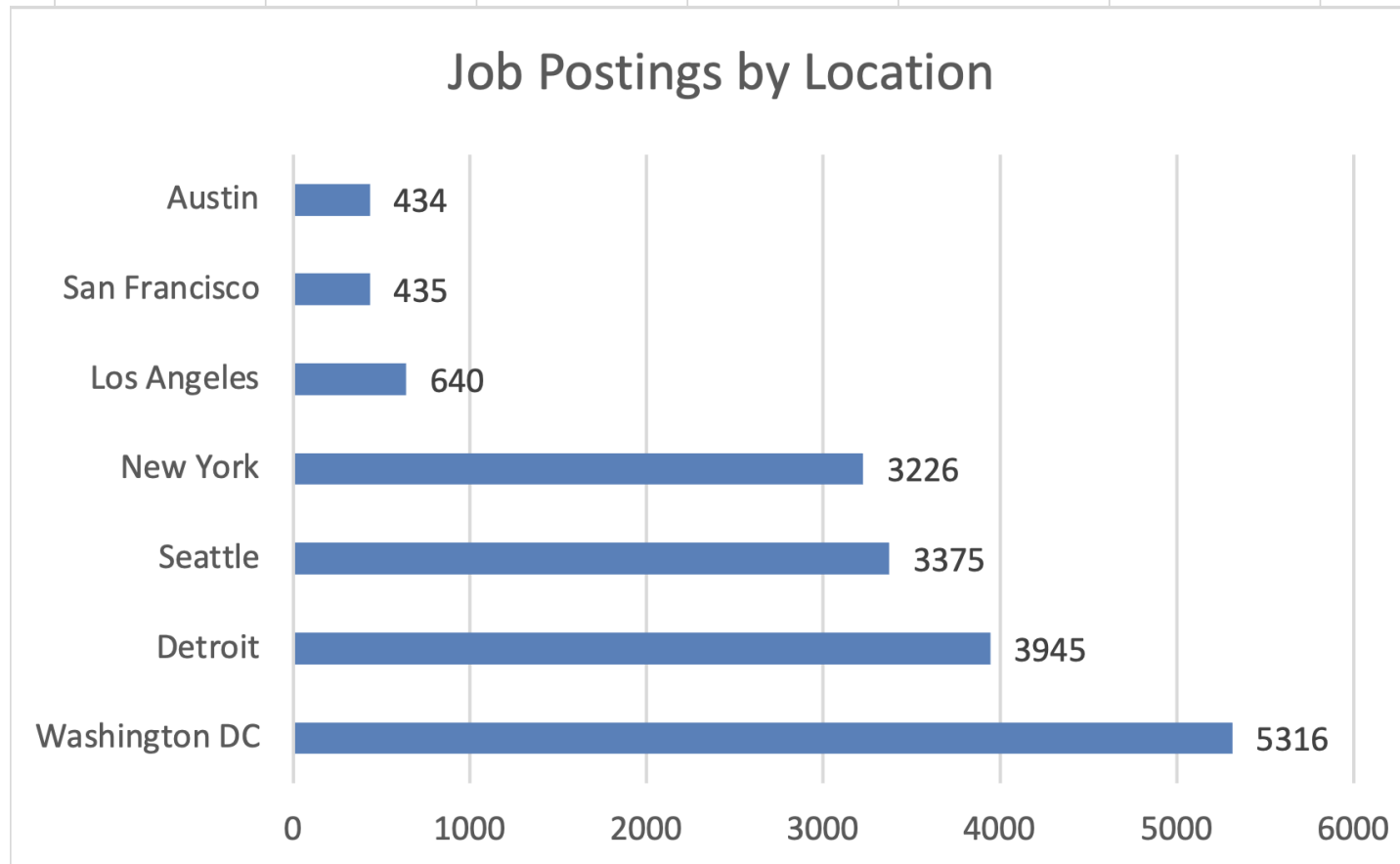
- **Data Source:** Stack Overflow Developer Survey dataset.
- **Data Collection:** Data was gathered using APIs, web scraping, and CSV files.
- **Data Wrangling:** Cleaned the dataset by handling missing values, removing inconsistencies, and preparing it for analysis.
- **Data Analysis Tools:**
 - Python (Pandas, Requests, BeautifulSoup)
 - SQL
 - IBM Cognos Analytics
- **Output:** Created visualizations, dashboards, and extracted meaningful insights from the survey data.



Results



JOB POSTINGS

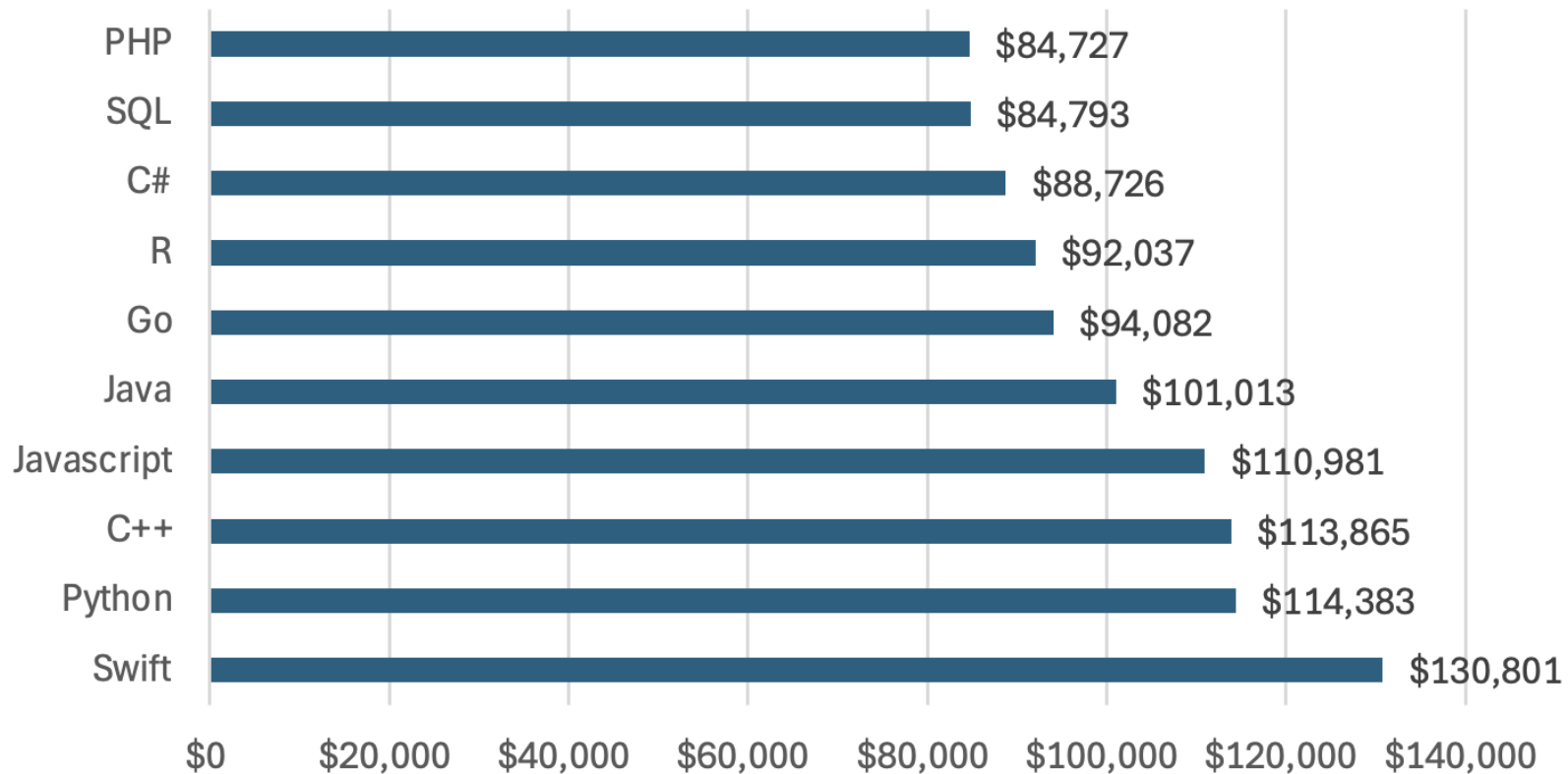


Observation:

Washington DC had the highest number of job postings (5,316), while Austin had the fewest (434) among the analyzed locations..

POPULAR LANGUAGES

Average Annual Salary



Observation:

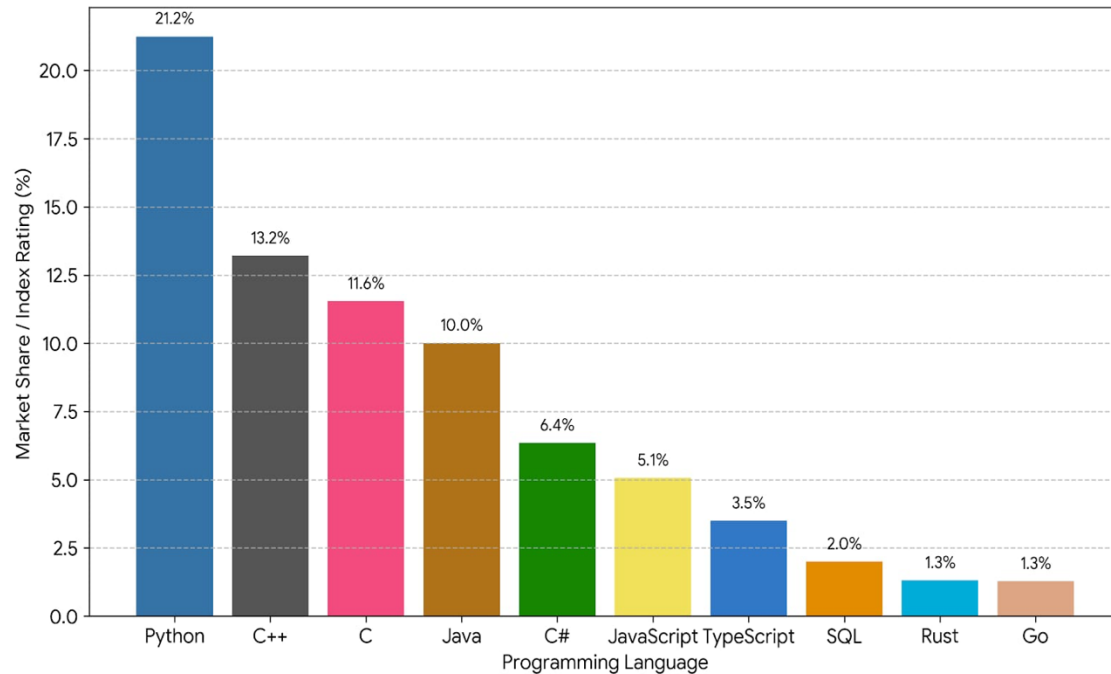
- Swift offers the highest average annual salary, followed by Python and C++. Java and Go also provide competitive salaries, while PHP and SQL have the lowest average salaries among the listed programming languages.



PROGRAMMING LANGUAGE TRENDS

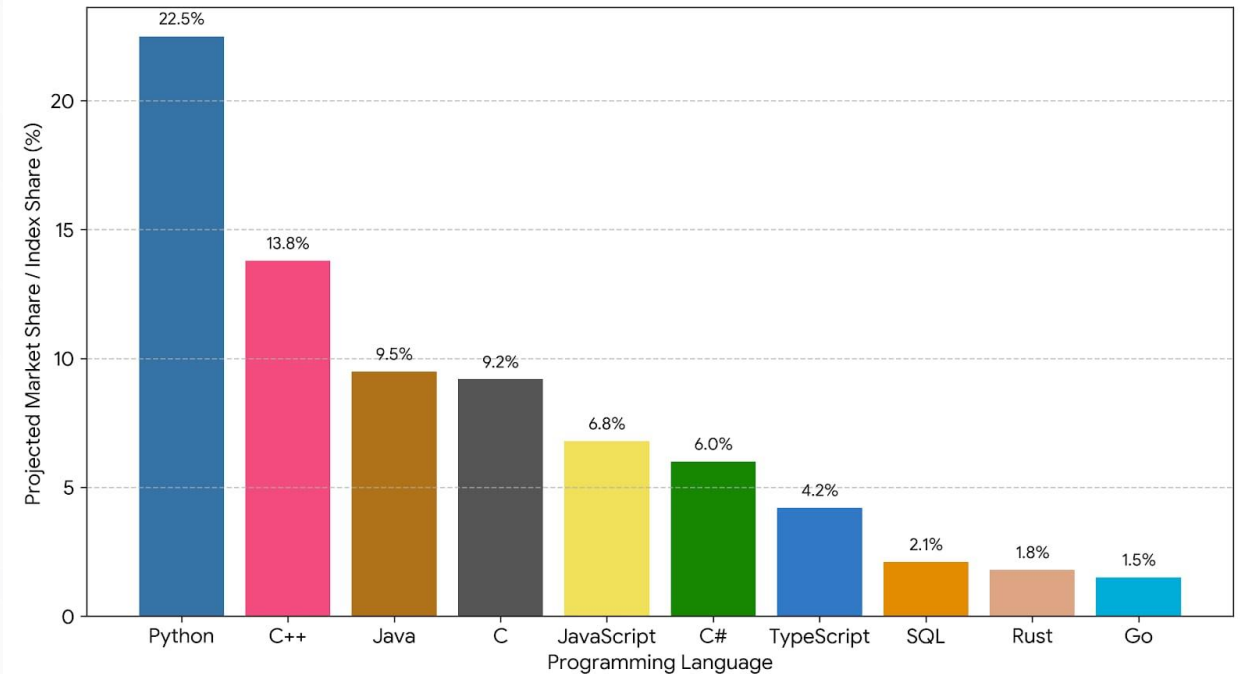
Current Year

Top 10 Programming Languages (2026 Index Shares)



Next Year

Projected Top 10 Programming Languages (2027 Outlook)



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Python jumps from *21.2% to 22.5%* market share, expanding its massive lead.
- C++ rises to *13.8%* while pure C drops to *9.2%*, reversing their traditional hierarchy.
- TypeScript climbs to *4.2%* and Rust grows to *1.8%*, showing the fastest upward velocity among newer languages.

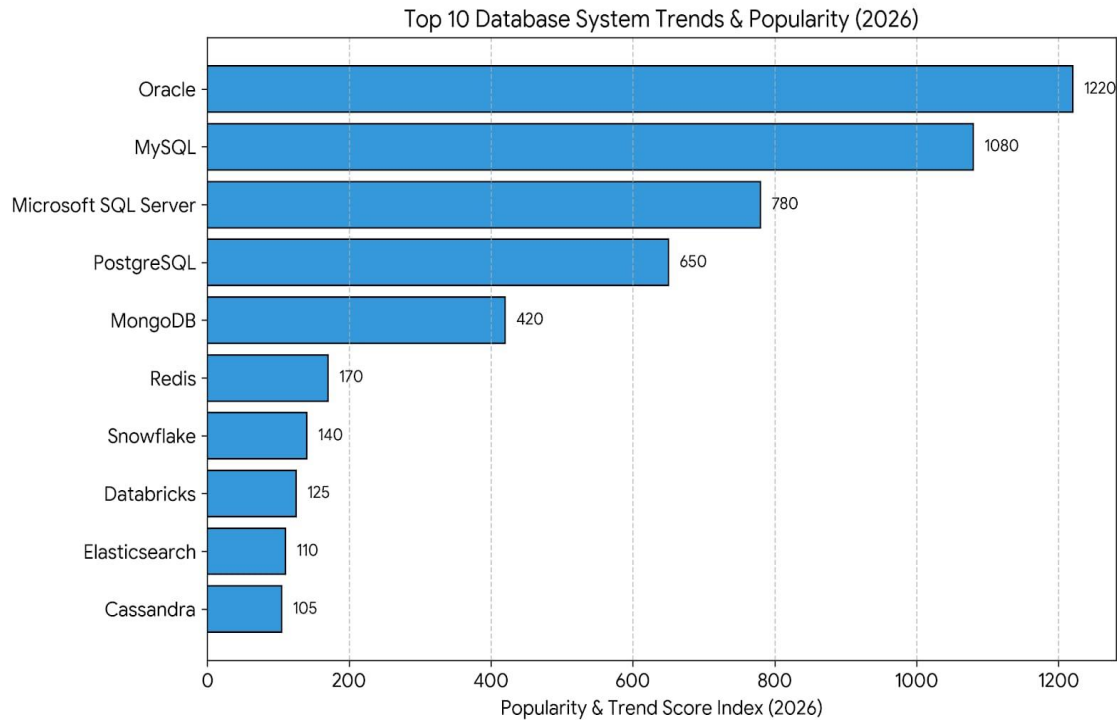
Implications

- Python has evolved from a general-purpose tool into the non-negotiable operating layer for artificial intelligence and LLM infrastructure.
- Compute-heavy industries like AI hardware acceleration, graphics, and high-frequency trading are aggressively prioritizing C++ abstractions over legacy C.
- Enterprises are actively restructuring their stacks around type-safe and memory-safe alternatives to meet modern security mandates and slash debugging overhead.

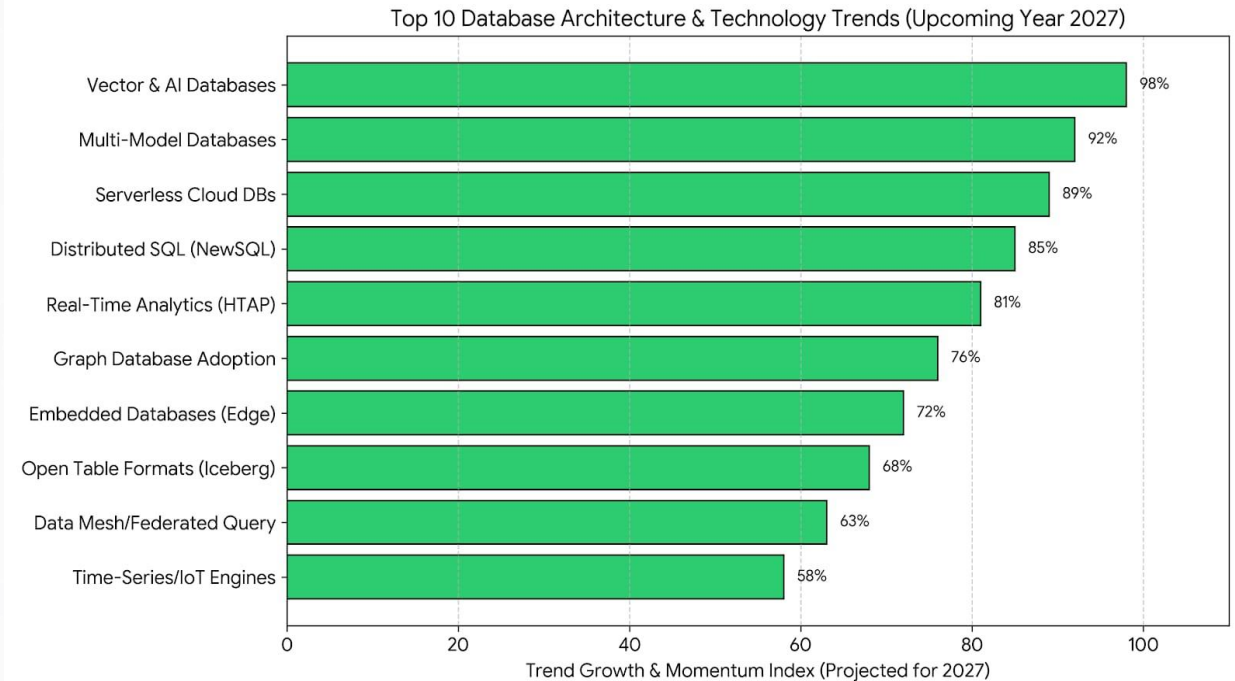


DATABASE TRENDS

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- ***AI Native:*** Vector capabilities are now mandatory for database engines to power AI apps.
- ***Consolidation:*** Unified multi-model platforms are replacing complex, fragmented multi-database stacks.
- ***Real-Time:*** Slow, batch ETL pipelines are dying in favor of instant, real-time analytics.

Implications

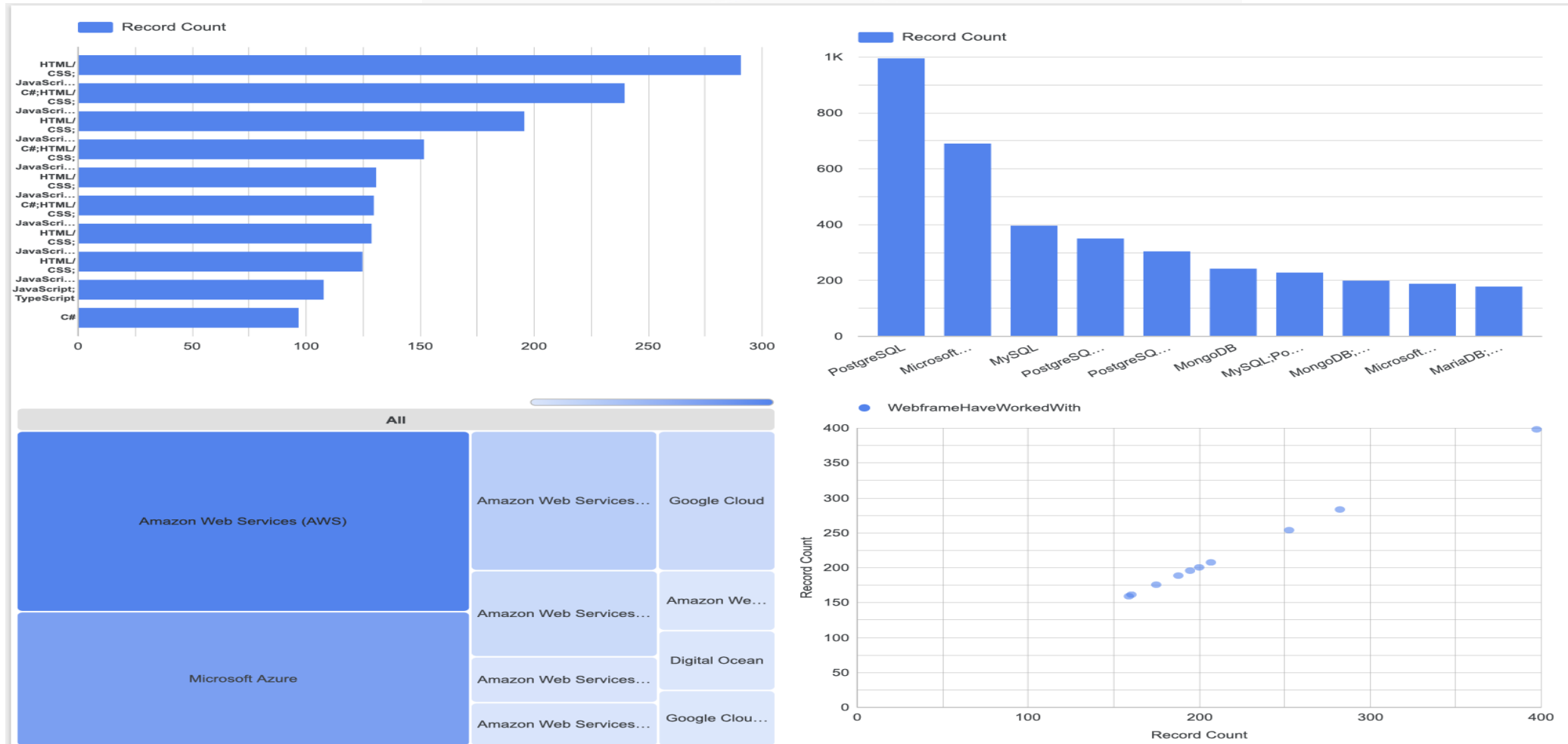
- ***Cut Complexity:*** Standardize on versatile, extensible engines to minimize operational overhead.
- ***Zero Lock-In:*** Adopt open, decoupled data formats to switch query tools freely.
- ***Reduce Costs:*** Migrate to serverless models to pay only for active compute.



DASHBOARD



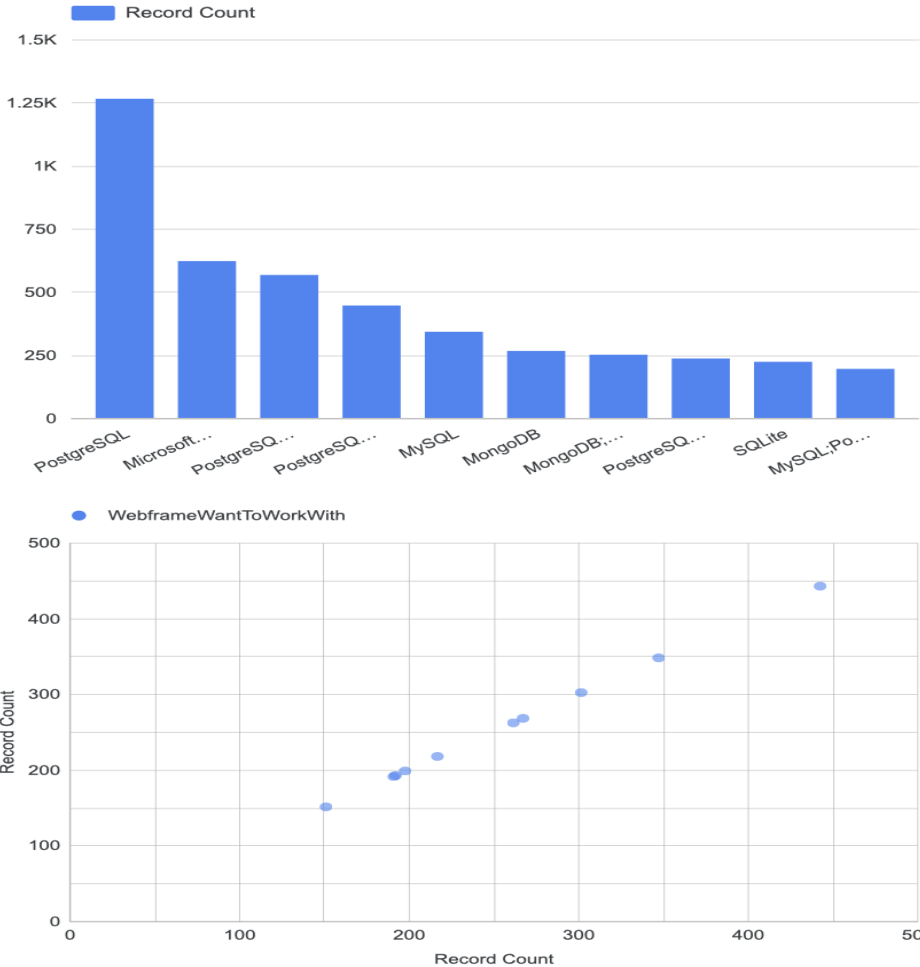
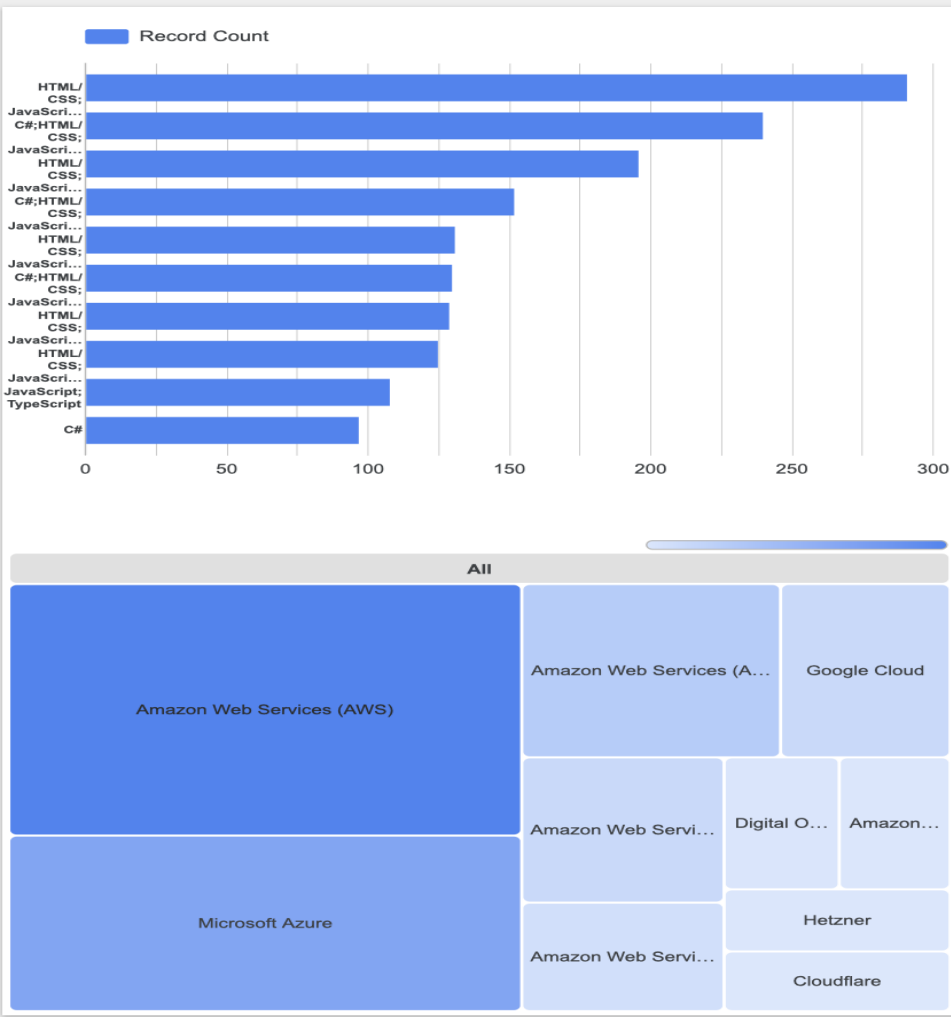
DASHBOARD TAB 1: Current Technology Usage



Key Insights of Dashboard TAB 1: Current Technology Usage

- **HTML/CSS, JavaScript, and C#** are the most commonly used technologies among survey respondents.
- **PostgreSQL** is the most widely used database, followed by **Microsoft SQL Server** and **MySQL**.
- **Amazon Web Services (AWS)** is the leading cloud platform, with **Microsoft Azure** and **Google Cloud** also showing strong adoption.
- The visualizations indicate that developers predominantly use established web technologies, relational databases, and major cloud platforms.

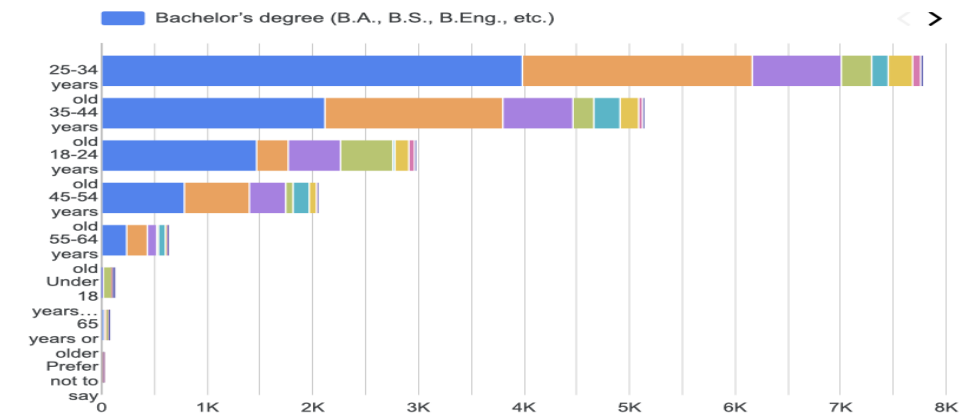
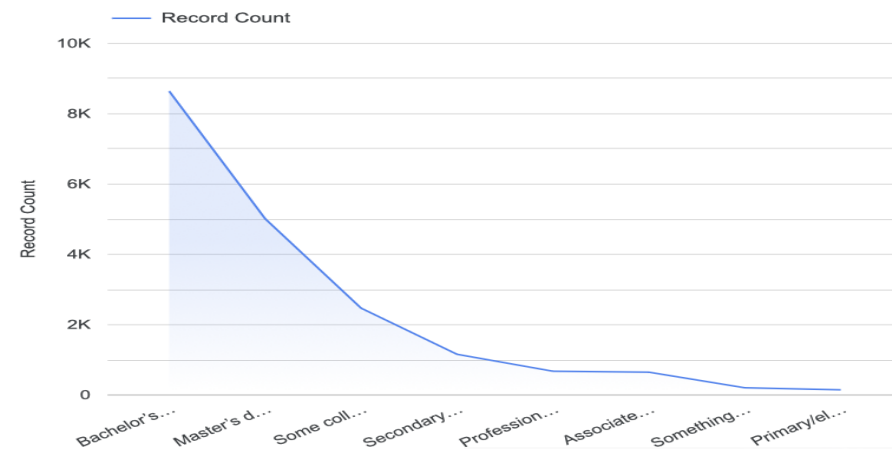
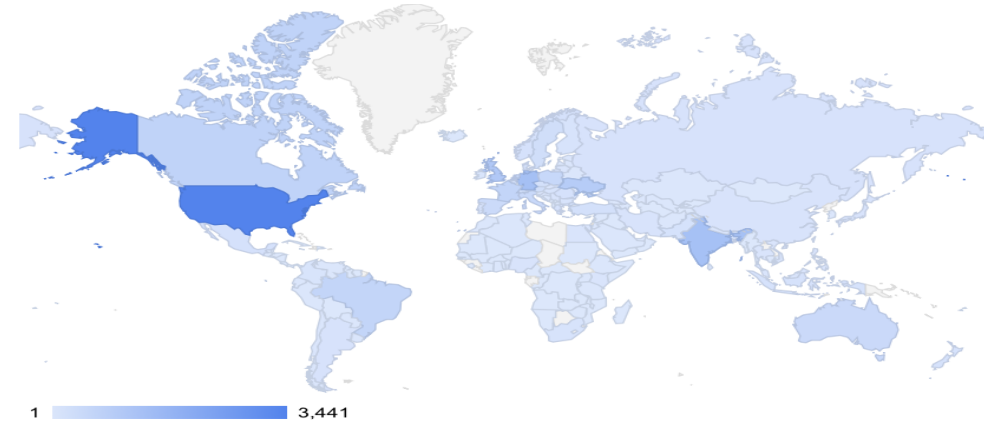
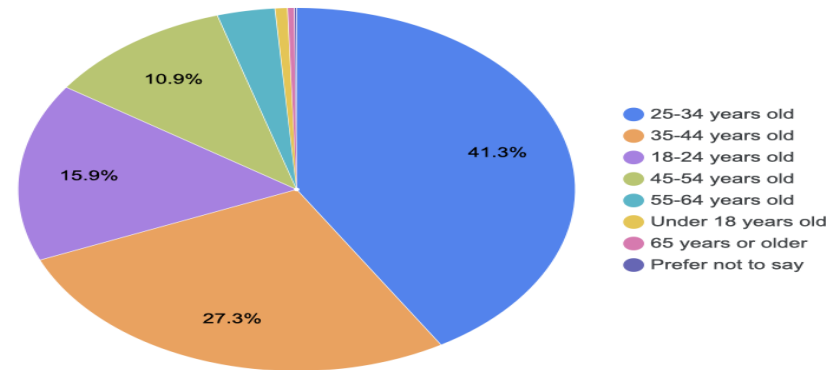
DASHBOARD TAB 2: Future Technology Trends



Key Insights of Dashboard TAB 2:Future Technology Trends

- **HTML/CSS, JavaScript, and C#** remain the most popular technologies currently used by developers.
- **PostgreSQL** is the most preferred database, with **Microsoft SQL Server** and **MySQL** also widely adopted.
- **Amazon Web Services (AWS)** is the leading cloud platform developers want to work with, followed by **Microsoft Azure** and **Google Cloud**.
- Overall, the trends indicate continued demand for **web development technologies, SQL databases, and cloud computing platforms**, highlighting their importance in the software industry.

DASHBOARD TAB 3: Demographics



Key Insights of Dashboard TAB 3: Demographics

- The **25–34 years** age group forms the largest share of respondents (**41.3%**), followed by **35–44 years (27.3%)**, indicating a predominantly early-career workforce.
- The **United States** has the highest number of survey respondents, with participation from developers across many countries worldwide.
- Most respondents hold a **Bachelor's degree**, making it the most common educational qualification among developers.
- The distribution across age and education suggests that **young professionals with undergraduate degrees make up the majority of the developer community**, while participation decreases in older age groups.



DISCUSSION



- The analysis highlights strong demand for **web technologies**, especially **HTML/CSS, JavaScript, and SQL-based databases**.
- **AWS, Microsoft Azure, and Google Cloud** are the most widely used and preferred cloud platforms, reflecting the growing importance of cloud computing.
- **Python, JavaScript, and modern web frameworks** continue to be among the most popular technologies for developers.
- Most survey respondents are **25–44 years old**, hold a **Bachelor's degree**, and are primarily from the **United States**, representing an experienced global developer community.
- Overall, the dashboards indicate that **cloud computing, web development, and data-related technologies** are expected to remain in high demand and provide strong career opportunities.

OVERALL FINDINGS & IMPLICATIONS

Findings

- Web technologies and cloud platforms dominate the current developer ecosystem.
- AWS, Microsoft Azure, and PostgreSQL are among the most widely used technologies.
- Most respondents are young professionals (25–44 years) with a Bachelor's degree.

Implications

- Learning cloud computing, SQL, and web development can improve career opportunities.
- Developers should focus on in-demand technologies such as Python, JavaScript, and cloud platforms.
- Organizations should invest in modern cloud and web technologies to meet future industry demands.



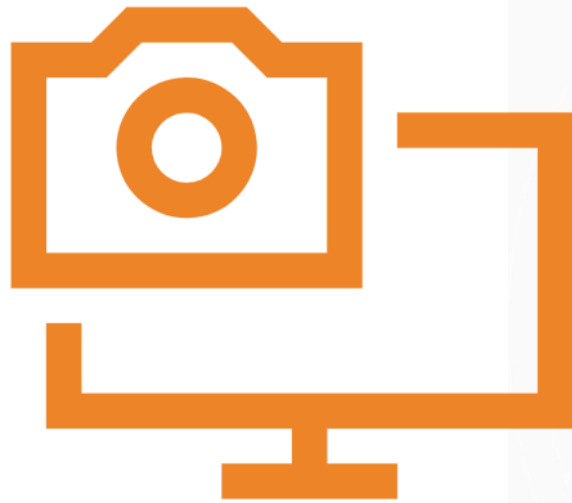
CONCLUSION



- The project successfully analyzed developer survey data using Excel, Python, and IBM Cognos Analytics.
- Interactive dashboards helped identify trends in programming languages, databases, cloud platforms, web technologies, and developer demographics.
- The analysis shows that cloud computing, web development, and data-related technologies continue to dominate the software industry.
- The insights gained can support students, developers, and organizations in making informed technology and career decisions.



APPENDIX



Resources Used

- Stack Overflow Developer Survey 2023 Dataset
- GOOGLE Looker Studio
- Microsoft Excel
- Python (Pandas, Matplotlib)
- IBM Skills Network Labs

